

# PROJECT DOCUMENTATION

## National Family Health Survey (NFHS) Data Analysis using Python

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### 1. Project Overview

This project analyzes National Family Health Survey (NFHS) data using Python to explore state-wise health and demographic indicators. The project focuses on female literacy, vaccination coverage, child nutrition, antenatal care, institutional births, and anaemia among women. Data preprocessing, exploratory data analysis, statistical techniques, and visualizations were performed to identify patterns and generate meaningful public health insights.

### 2. Tools & Technologies

| Tool             | Purpose                      |
|------------------|------------------------------|
| Python           | Programming                  |
| Pandas           | Data Cleaning & Manipulation |
| NumPy            | Numerical Operations         |
| Matplotlib       | Visualization                |
| Seaborn          | Statistical Visualization    |
| Jupyter Notebook | Development Environment      |

### 3. Dataset

Source: National Family Health Survey (NFHS). The dataset contains state-level health indicators covering maternal healthcare, child health, literacy, nutrition, immunization, and related demographic information used for comprehensive exploratory analysis.

### 4. Project Workflow

Data Collection → Data Cleaning & Preprocessing → Exploratory Data Analysis → GroupBy & Pivot Analysis → Correlation Analysis → 11 Visualizations → Insights & Conclusion

### 5. Key Findings

- Female literacy shows a positive relationship with maternal healthcare and vaccination.
- Institutional births are generally higher in better-performing states.
- Child stunting varies considerably across states.
- Anaemia among women remains a significant health concern.
- Correlation analysis revealed meaningful relationships among major health indicators.
- State-wise comparisons highlighted regional disparities.

### 6. Visualizations

The project includes 11 visualizations developed using Matplotlib and Seaborn, including histograms, bar charts, box plots, heatmaps, and comparative state-wise charts illustrating literacy, vaccination, child nutrition, maternal healthcare, institutional births, anaemia, and other key indicators.

### 7. Files Included

- National Family Health Survey Analysis.ipynb
- NFHS Dataset (.csv)
- README.md

- Project Documentation.pdf

## 8. How to Use

1. Open the notebook in Jupyter Notebook.
2. Install Pandas, NumPy, Matplotlib, and Seaborn.
3. Place the dataset in the project folder.
4. Run all notebook cells sequentially.
5. Review the analyses, visualizations, and observations.

## 9. Conclusion

This project demonstrates the practical application of Python-based data analytics techniques on a real-world public health dataset. Through systematic preprocessing, exploratory analysis, and 11 insightful visualizations, the project highlights regional trends and relationships among important health indicators, supporting evidence-based decision-making.